

Hidden Markov Model with Student's t Emissions for Regimes of a Bank Bond Portfolio Return

Analytical Report

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Abstract

We apply a multivariate hidden Markov model (HMM) with multivariate Student's t-distribution emissions to daily *total* returns of a bond portfolio, aggregated into seven segments of the Moscow Exchange (FLB of three duration segments, corporate issues of three listing tiers, and mortgage bonds). Total return accounts for both the change in clean price and coupon income. The model is trained via the ECM method with a warm start from a Gaussian HMM on a sample of **904** trading days (July 2021 – July 2026). Model selection by BIC over the number of regimes yields $K = 4$; the likelihood ratio test rejects Gaussian emissions in favor of Student's t emissions for all K ($p < 0.001$). The estimated degrees of freedom $\nu \in [2.9, 14.2]$ directly measure the tail heaviness of each regime. In a strict *rolling out-of-sample* backtest with a one-day-ahead forecast, the unconditional normal approximation is rejected by the Kupiec and Christoffersen tests at the 99% level (realized breach frequency 2.48% versus the nominal 1%), and its Expected Shortfall statistically understates risk (Acerbi-Szekely test, $p = 0.043$); the t-HMM, however, is correctly calibrated on both measures and is not inferior to specialized tail models — univariate EVT-GPD and GARCH- t , as well as multivariate DCC-GARCH — while at the same time providing an interpretable regime structure that the alternatives lack. Block bootstrap quantitatively assesses parameter uncertainty, Euler decomposition attributes risk to portfolio segments, and a reduced-form transition logit identifies the CBR key rate as the main macro driver of entry into stress, producing an early warning signal (AUC 0.88).

Keywords: hidden Markov models, Student's t-distribution, regime switching, heavy tails, VaR, Expected Shortfall, backtest, EVT, GARCH, bonds, FLB.

1 Introduction

Fixed income returns exhibit *regime switching*: an alternation of calm periods with moderate volatility and episodes of stress with sharp price movements, volatility clustering, and heavy tails. For a bank's bond portfolio, identifying such regimes is important for managing interest rate and market risk, stress testing, and computing regime-conditional risk measures (VaR, Expected Shortfall).

Classical regime-switching models (Hamilton, 1989) use a hidden Markov chain governing the emission parameters. However, Gaussian emissions systematically underestimate the frequency of extreme movements. We replace them with a multivariate Student's t distribution: the degrees-of-freedom parameter ν_k of each regime directly measures the heaviness of its tails, and estimation via the ECM method (Liu & Rubin, 1995), through the representation as a Gaussian scale mixture, makes the fit robust to outliers (Bulla & Bulla, 2006).

Contribution of the work: (i) construction of a time-consistent multivariate panel of *full* returns of a real bond portfolio based on Moscow Exchange data; (ii) BIC-justified selection of the number

of regimes and a statistical test of Student’s *t* emissions against Gaussian ones; (iii) regime-conditional risk measures; (iv) a rigorous rolling out-of-sample VaR backtest with Kupiec and Christoffersen tests and comparison with benchmark tail-risk models (EVT-GPD and GARCH-*t*).

2 Data

The original portfolio contains **178** issues. Of these, **16** are eurobonds (ISIN with prefix **XS/CH**), not traded on the Moscow Exchange and therefore excluded. Of the **162** ruble-denominated issues, **130** were identified on the MOEX platform (the remaining ones are over-the-counter placements or issues redeemed outside the ISS archive), of which **124** have a liquid daily quote history.

2.1 Total return

Daily quotes were downloaded via the public API of the Moscow Exchange Information and Statistics Server (ISS). Unlike the preliminary analysis on clean prices, we compute the *total* return of each issue by adding the coupon income:

$$r_t = \frac{P_t^{\text{clean}} - P_{t-1}^{\text{clean}}}{P_{t-1}^{\text{clean}}} + \frac{c/100}{365}, \quad (1)$$

where P^{clean} is the clean price, c is the issue’s annual coupon rate (field **COUPONPERCENT**). The second term is the daily coupon carry. This approach is robust to noise in the accrued coupon income field (**ACCINT**), which for some issues in the ISS archive contains artifacts. Moves exceeding 15% in a day are discarded as erroneous quotes. The switch from clean price to total return shifts all segment drifts from the negative region (the effect of price amortization to par) into the positive region, while volatility and tail heaviness remain essentially unchanged.

The issues are aggregated into **7** segments by instrument type, duration (for FLB — by time to maturity at the end of the sample) and credit quality (for corporate bonds — by MOEX listing level as a proxy):

Table 1: Portfolio segments and number of issues.

Segment	Code	Issues
FLB short (<3 yrs)	FLB_short	13
FLB medium (3–7 yrs)	FLB_med	15
FLB long (≥7 yrs)	FLB_long	20
Corporate, listing 1	Corp_L1	16
Corporate, listing 2	Corp_L2	17
Corporate, listing 3	Corp_L3	22
Mortgage-backed (DOM.RF Mortgage Agent)	MBS	21

The segment’s daily return index is the equal-weighted average of the total returns of all its issues traded on that day (provided at least three bonds are present), which makes the segment robust to the entry and exit of individual issues. The final panel covers **904** trading days (2021-07-29 – 2026-07-17) across 7 segments. The portfolio as a whole is defined as an equal-weighted combination of segments and yields approximately 5% annualized at an annual volatility of 4.3%. All series exhibit excess kurtosis (from 2.5 to 20), which motivates the use of Student’s *t* emissions.

3 Methodology

3.1 Model

Let $\mathbf{x}_t \in \mathbb{R}^d$ be the vector of returns of $d = 7$ segments on day t , and $s_t \in \{1, \dots, K\}$ be the hidden regime, forming a Markov chain with transition matrix $\mathbf{A} = (a_{ij})$, $a_{ij} = P(s_{t+1} = j \mid s_t = i)$. Conditional on regime k , the observation has a multivariate Student distribution:

$$\mathbf{x}_t \mid s_t = k \sim t_{\nu_k}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad (2)$$

with location vector $\boldsymbol{\mu}_k$, scale matrix $\boldsymbol{\Sigma}_k$, and degrees of freedom $\nu_k > 2$. The density admits a representation as a Gaussian scale mixture: $\mathbf{x}_t \mid s_t = k, u \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k/u)$, where $u \sim \text{Gamma}(\nu_k/2, \nu_k/2)$. A small ν_k (on the order of 3–5) implies heavy tails and frequent extremes; $\nu_k > 30$ is practically indistinguishable from the Gaussian case.

3.2 Estimation

Parameters are estimated by the ECM (Expectation–Conditional–Maximization) method. In the E-step, the forward–backward algorithm yields smoothed regime probabilities $\gamma_t(k) = P(s_t = k \mid \mathbf{x}_{1:T})$, and for the Student component — the conditional expectation of the latent scale $E[u \mid \mathbf{x}_t, s_t = k] = (\nu_k + d)/(\nu_k + \delta_t^2)$, where $\delta_t^2 = (\mathbf{x}_t - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_t - \boldsymbol{\mu}_k)$ is the squared Mahalanobis distance. This weight automatically downweights the influence of outliers when estimating $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$, ensuring robustness. The degrees of freedom ν_k are updated by solving a one-dimensional equation; the lower bound $\nu \geq 2.5$ prevents a single regime from degenerating into an "outlier trap." The data are standardized before fitting; the estimates are returned in the original units. For reliable convergence, a warm start from a Gaussian HMM is used. Among the local optima, we select the ergodic (non-absorbing) solution with a proper stationary distribution and finite regime durations; it is statistically equivalent to the global one (log-likelihood difference < 5).

3.3 Choice of the number of regimes

The number of regimes K is chosen using the Bayesian information criterion (BIC). Since the Gaussian HMM is the limiting case $\nu_k \rightarrow \infty$, the likelihood ratio (LR) test with K degrees of freedom is valid for the hypothesis "Student vs. Gaussian."

Table 2: Model selection. The Student HMM minimizes BIC at $K = 4$ and dominates the Gaussian one for all K (LR test, $\text{df} = K$).

K	ℓ_t	BIC_t	ℓ_g	BIC_g	LR	p
2	-7183.4	14877.3	-7724.2	15945.2	1081.5	< 0.001
3	-7051.9	14893.4	-7281.9	15333.0	460.0	< 0.001
4	-6834.4	14751.0	-7008.2	15071.5	347.7	< 0.001
5	-6728.6	14845.8	-6878.5	15111.5	299.7	< 0.001

4 Results

4.1 Regime structure

The model with $K = 4$ identifies four regimes with clear economic interpretation (table 3). Regimes are ordered by portfolio return volatility. The key difference between regimes is not only the level of volatility, but also the *tail heaviness* ν , the *duration*, and the *sign of the drift*.

Table 3: Regime parameters. ν — degrees of freedom (tail heaviness); portfolio volatility — annualized; mean drift — in basis points per day (including coupon carry); duration — expected number of days in the regime; stationary and Viterbi shares.

Regime	ν	Vol. %	Drift, bp	Self-pers.	Dur., days	Station. %	Viterbi %
R1 (quiet, heavy tails)	2.88	2.96	+1.0	0.752	4.0	14.4	22.6
R2 (sell-off)	5.19	3.80	-1.2	0.819	5.5	20.5	30.4
R3 (sustained rally)	14.18	4.01	+3.0	0.974	38.8	43.1	30.1
R4 (high volatility)	7.94	6.18	+4.8	0.943	17.5	22.1	16.9

- **R1 (quiet, heavy tails, $\nu = 2.9$):** the lowest volatility (3.0%), but the heaviest tails in the sample — jump risk is present even in calm periods. Short episodes (expected duration 4 days).
- **R2 (sell-off, $\nu = 5.2$):** the only regime with negative drift (−1.2 bp/day), moderately heavy tails; hits long FLB and mortgage bonds hardest (see below).
- **R3 (sustained rally, $\nu = 14$):** the dominant regime (43.1% of stationary mass), positive drift, near-Gaussian tails, and high self-persistence (0.974, duration ≈ 39 days). Corresponds to the rally of 2025–2026 amid the policy rate cut.
- **R4 (high volatility, $\nu = 7.9$):** the highest volatility (6.2%), but, notably, a *positive* drift (+4.8 bp/day) — this is the volatile recovery of late 2024 – early 2025, not stress alone.

Figure 1 shows the normalized portfolio value colored by the decoded (Viterbi) regime and the smoothed posterior regime probabilities. Including coupon income, the portfolio gains about 20% over the period. The timeline is consistent with events in the Russian debt market: a calm start (R1), a prolonged 2023–2024 sell-off (R2), a volatile bottom in late 2024 (R4), and a sustained 2025–2026 rally (R3).

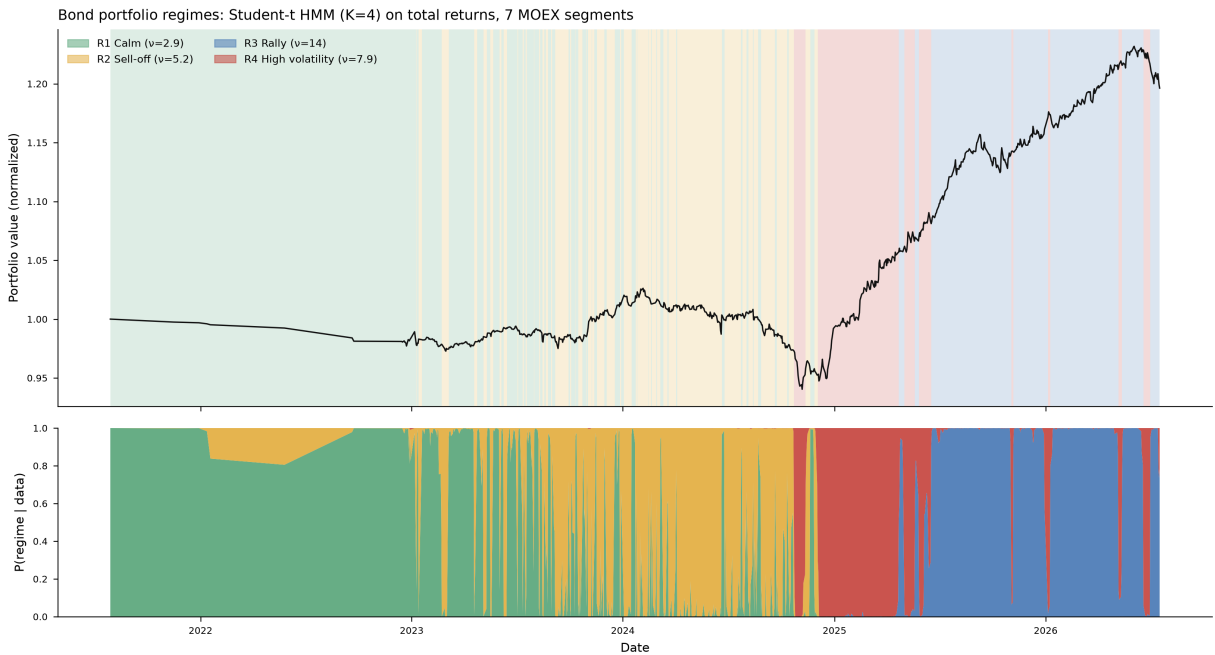


Figure 1: **Top:** normalized portfolio value on a total-return basis; background colored by the decoded Viterbi regime. **Bottom:** smoothed posterior regime probabilities. The portfolio is an equal-weighted combination of 7 segments, 904 trading days.

4.2 Transition dynamics and segment profile

The transition matrix (figure 2, left; table 4) confirms the high persistence of regimes R3 and R4. The heatmap of mean segment returns (right) shows that the R2 sell-off hits long FLB hardest (-4.6 bp/day) and mortgage bonds (-3.6), whereas in regimes R3 and R4 almost all segments deliver positive returns, with first-tier corporate issues in R4 giving the largest drift ($+10.7$ bp/day).

Table 4: Transition matrix $\hat{\mathbf{A}}$ (row — from, column — to).

	R1	R2	R3	R4
R1	0.752	0.248	0.000	0.000
R2	0.174	0.819	0.000	0.007
R3	0.000	0.000	0.974	0.026
R4	0.000	0.007	0.050	0.943

Regime structure: R3/R4 are the most persistent; sell-off R2 hits long OFZ and MBS the hardest

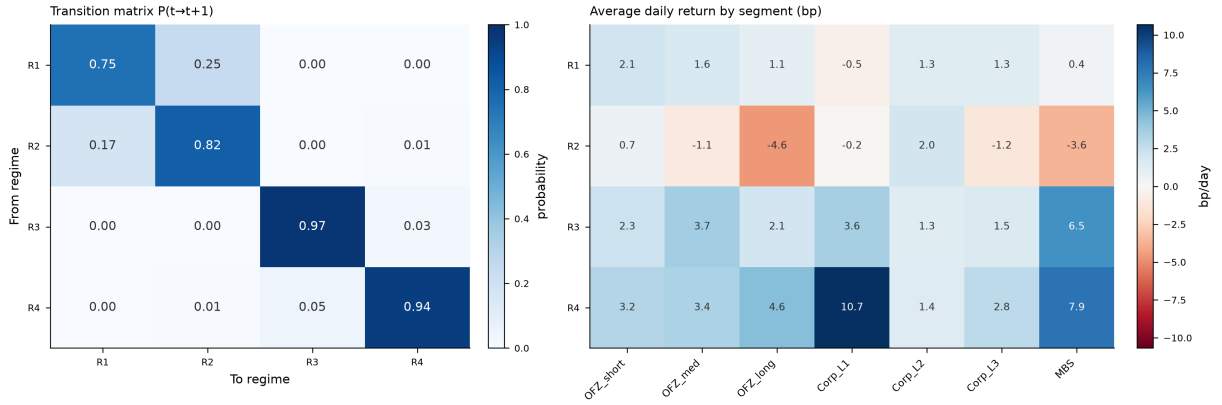


Figure 2: **Left:** transition matrix. **Right:** mean daily total return (bp) by segment in each regime; a diverging scale centered on zero.

Table 5: Mean daily total return by segment in each regime (basis points).

Regime	FLB_short	FLB_med	FLB_long	Corp_L1	Corp_L2	Corp_L3	MBS
R1	+2.1	+1.6	+1.1	-0.5	+1.3	+1.3	+0.4
R2	+0.7	-1.1	-4.6	-0.2	+2.0	-1.2	-3.6
R3	+2.3	+3.7	+2.1	+3.6	+1.3	+1.5	+6.5
R4	+3.2	+3.4	+4.6	+10.7	+1.4	+2.8	+7.9

4.3 Tail diagnostics

Figure 3 compares the empirical quantiles of portfolio returns in each regime with the theoretical quantiles of the fitted Student distribution and a normal approximation. In regime R1 ($\nu = 2.9$), the Student quantiles accurately reproduce the extreme observations, whereas the normal quantiles are compressed and underestimate the tail; in the near-Gaussian regime R3 ($\nu = 14$), both approximations coincide.

QQ tails of portfolio returns: Student-t fits the tails, Normal does not

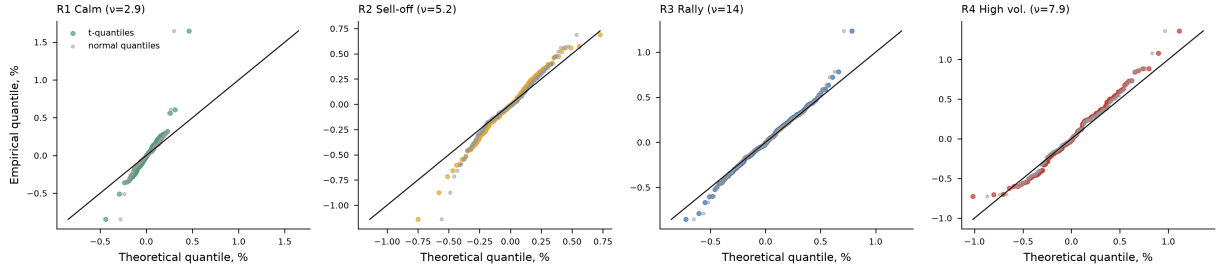


Figure 3: QQ plots of portfolio returns by regime: empirical quantiles versus Student (colored) and normal (gray) quantiles.

5 Risk measures by regime

For each regime, the 95% and 99% Value-at-Risk (VaR) and Expected Shortfall (ES) of portfolio returns are computed — empirically and parametrically from a fitted Student distribution (table 6, figure 4). ES is noticeably deeper than VaR in all regimes; the smallest tail risk is carried by the quiet regime R1, while the largest at the 99% level belongs to the sell-off regime R2 (despite not being the highest in volatility, its tails are heavier than those of R3/R4).

Table 6: Regime-conditional VaR and ES of portfolio returns (% per day; negative values — losses). Empirical and parametric (Student) estimates.

Regime	Level	VaR emp.	ES emp.	VaR t	ES t
R1 (quiet, heavy tails)	95%	-0.243	-0.365	-0.237	-0.404
R1 (quiet, heavy tails)	99%	-0.357	-0.571	-0.474	-0.751
R2 (sell-off)	95%	-0.416	-0.600	-0.387	-0.546
R2 (sell-off)	99%	-0.671	-0.911	-0.634	-0.829
R3 (steady rally)	95%	-0.378	-0.535	-0.382	-0.508
R3 (steady rally)	99%	-0.621	-0.770	-0.584	-0.702
R4 (high volatility)	95%	-0.551	-0.635	-0.578	-0.799
R4 (high volatility)	99%	-0.701	-0.714	-0.929	-1.164

Regime-conditional risk: ES is markedly deeper than VaR; R1 carries the lowest tail risk, R2 the highest at 99%

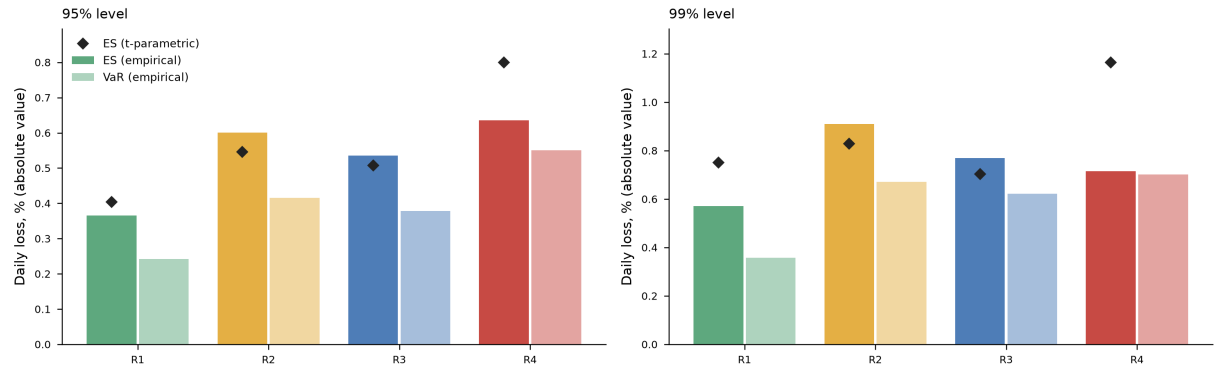


Figure 4: Regime-conditional ES (saturated) and VaR (light) by regime; diamonds — parametric Student ES.

6 Rolling Out-of-Sample Backtest

In-sample goodness of fit does not guarantee the quality of risk forecasts. We therefore conduct a rigorous *rolling* backtest with a one-day-ahead forecast. On each trading day t (starting from the 500th observation) the model is re-estimated on the entire available history $\mathbf{x}_{1:t}$ (re-estimation every 5 days), the *filtered* regime probability $P(s_t | \mathbf{x}_{1:t})$ is computed by a direct filter, the state distribution at $t + 1$ is forecast via the transition matrix, and VaR and ES are extracted from the corresponding mixture of Student portfolio densities. The forecast is then compared against the realized return on day $t + 1$. The resulting 403 out-of-sample forecasts are checked with the Kupiec test for unconditional coverage (the proportion of breaches) and the Christoffersen test for conditional coverage (which accounts for clustering of breaches).

We compare t-HMM with four benchmarks on the same days: the unconditional normal approximation (*Gaussian*), rolling historical simulation (*Empirical*), extreme value theory via the peaks-over-threshold method with a generalized Pareto distribution (*EVT-GPD*), and a univariate GARCH(1,1) with Student innovations (*GARCH-t*).

Table 7: VaR coverage backtest (403 out-of-sample days). Number of breaches, their share, and p -values of the Kupiec test (unconditional coverage) and Christoffersen test (conditional coverage). A model is rejected at $p < 0.05$.

Level	Model	Breaches	Share	Kupiec p	Christ. CC p	Verdict
5%	t-HMM	20	4.96%	0.973	0.999	—
5%	Gaussian	24	5.96%	0.392	0.652	—
5%	Empirical	26	6.45%	0.200	0.377	—
5%	EVT-GPD	27	6.70%	0.136	0.266	—
5%	GARCH-t	19	4.71%	0.791	0.959	—
1%	t-HMM	2	0.50%	0.260	0.525	—
1%	Gaussian	10	2.48%	0.012	0.034	rejected
1%	Empirical	5	1.24%	0.640	0.842	—
1%	EVT-GPD	4	0.99%	0.988	0.960	—
1%	GARCH-t	2	0.50%	0.260	0.525	—

The result is unambiguous (Figures 5–6). At the 99% level the unconditional normal approximation is **rejected** by both the Kupiec test ($p = 0.012$) and the Christoffersen test ($p = 0.034$): the Gaussian VaR is breached 2.48 times more often than the nominal rate. All the heavy-tailed models (t-HMM, EVT-GPD, GARCH- t) and the historical simulation pass both tests. EVT-GPD, optimized specifically for the tail, shows nearly perfect coverage (0.99%); t-HMM and GARCH- t are somewhat conservative (0.50%). At the 95% level all models are adequate — the discrepancy shows up precisely in the far tail.

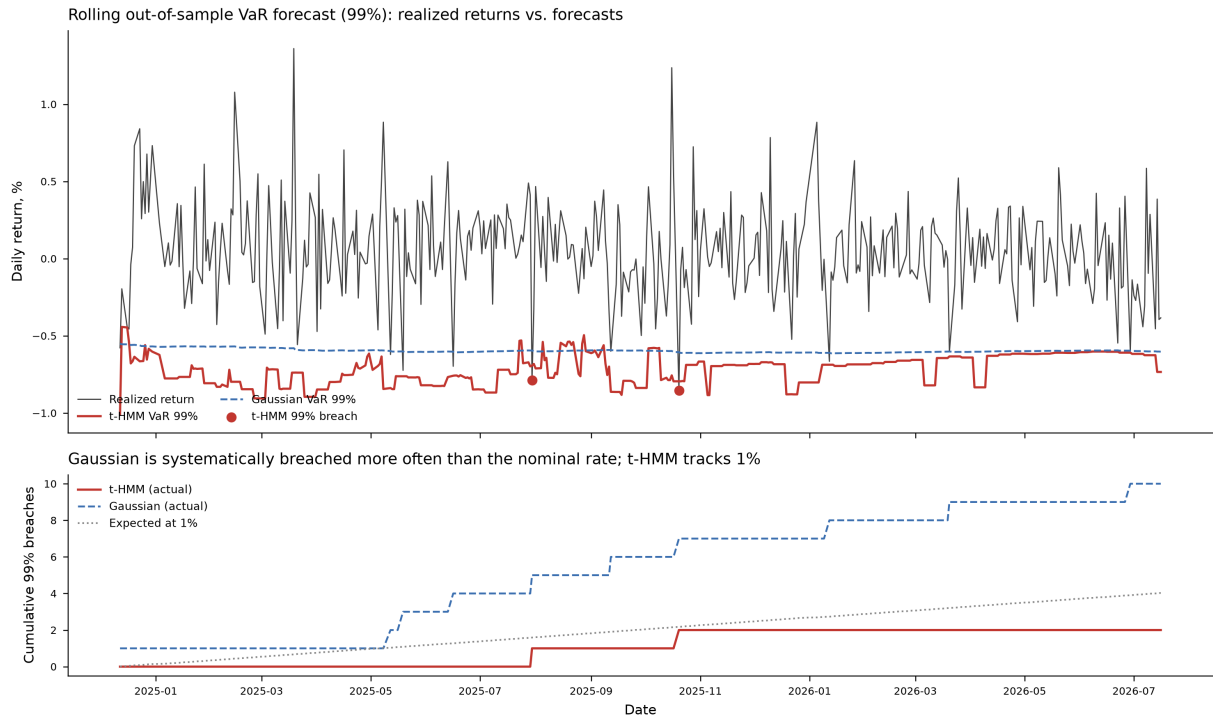


Figure 5: Top: realized daily return and rolling 99% VaR forecasts (t-HMM and Gaussian); dots mark t-HMM breaches. **Bottom:** cumulative number of breaches versus the expected number at 1%. Gaussian is systematically breached more often than the nominal rate.

VaR coverage backtest: at 99% only Gaussian is rejected; heavy-tailed models are well-calibrated

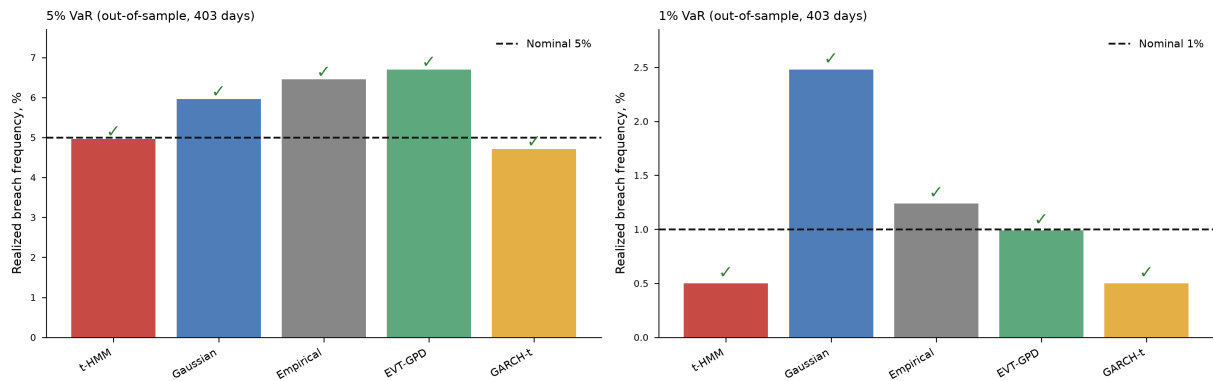


Figure 6: Realized VaR breach frequency across five models versus the nominal level (dashed line). A checkmark means the model passes the Kupiec test, a cross means it is rejected. At 99% only Gaussian is rejected.

The advantage of t-HMM over univariate EVT and GARCH lies not in tail accuracy (where they are comparable), but in the fact that t-HMM is the only one that provides an *interpretable multivariate regime structure*: it simultaneously reports which state the market is in, how long it will last, and how risk is distributed across the portfolio segments. EVT and GARCH do not provide this information.

7 Expected Shortfall Backtest

The Kupiec and Christoffersen tests check VaR coverage, but the regulatory measure (FRTB) is Expected Shortfall. We check ES calibration with the direct Acerbi–Szekely test (Test 2): the statistic $Z_2 = \frac{1}{T\alpha} \sum_t \frac{X_t \mathbf{1}\{X_t < \text{VaR}_t\}}{\text{ES}_t} + 1$ has zero expectation under correct calibration; $Z_2 > 0$ means that the ES underestimates realized losses in the tail. p -values are obtained by bootstrapping the null distribution. The test is run on the same 403 out-of-sample days for all five univariate models.

Table 8: Expected Shortfall backtest (Acerbi–Szekely, Test 2). $Z_2 > 0$ — underestimation of tail risk; the model is rejected at $p < 0.05$.

Level	Model	Breaches	Z_2	p	Verdict
95%	EVT-GPD	27	+0.274	0.249	calibrated
95%	Empirical	26	+0.241	0.311	calibrated
95%	GARCH-t	19	-0.160	0.412	calibrated
95%	Gaussian	24	+0.273	0.278	calibrated
95%	t-HMM	20	-0.038	0.836	calibrated
99%	EVT-GPD	4	-0.081	0.750	calibrated
99%	Empirical	5	+0.136	0.861	calibrated
99%	GARCH-t	2	-0.570	0.105	calibrated
99%	Gaussian	10	+1.437	0.043	underestimates risk
99%	t-HMM	2	-0.562	0.102	calibrated

The result reinforces the VaR finding: at the 99% level **the Gaussian approximation’s ES statistically underestimates risk** ($Z_2 = +1.44$, $p = 0.043$) — not only is the Gaussian VaR breached too often, but the average magnitude of losses beyond the threshold is also underestimated. The ES of all heavy-tailed models (t-HMM, EVT-GPD, GARCH- t) and the historical simulation are calibrated (figure 7). This is direct confirmation of the practical value of Student’s- t emissions specifically for the risk measure adopted by Basel.

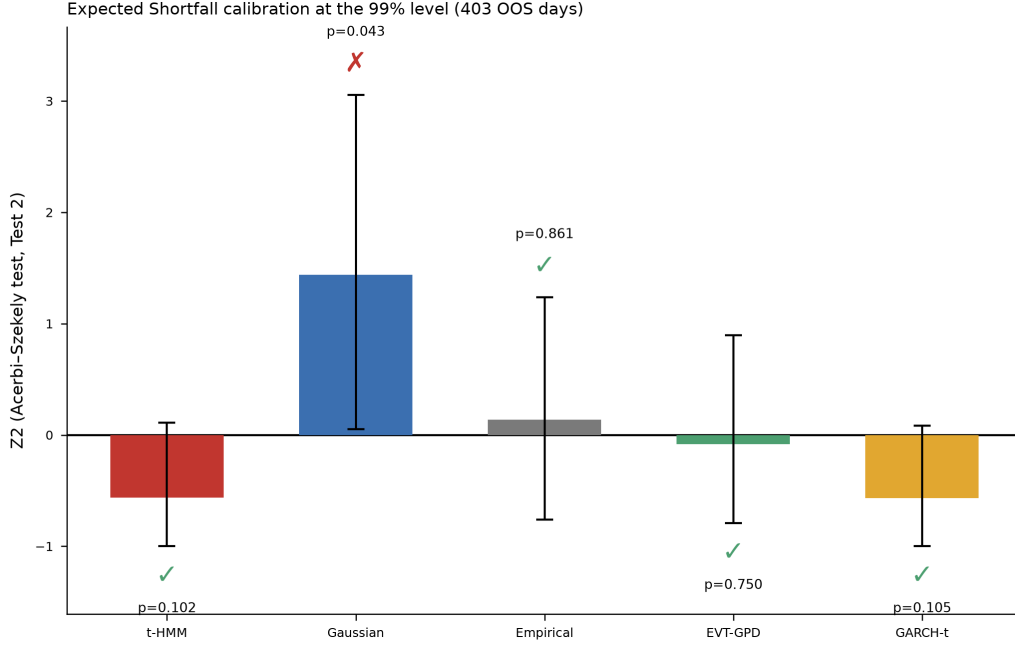


Figure 7: Acerbi-Szekely statistic Z_2 by model at the 99% level with bootstrap intervals. Bars above zero indicate underestimation of tail risk; a cross marks a model rejected at ($p < 0.05$).

8 Risk decomposition by segment

For portfolio management it is important to know not only the aggregate risk but also the contribution of each segment. For an elliptical (Student) portfolio with weights $\mathbf{w}$ and scale matrix Σ , VaR decomposes additively via Euler's theorem: the marginal contribution of segment i is proportional to $(\Sigma \mathbf{w})_i / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$, and the component contribution $\text{CVaR}_i = w_i \cdot (\text{marginal})_i$ sums to the portfolio VaR. We present the decomposition for an equally-weighted portfolio; the formula accepts *any* weight vector, so switching to the bank's actual weights reduces to substituting $\mathbf{w}$.

Table 9: Unconditional decomposition of 99% portfolio VaR and ES by segment (contribution shares; equal weights). Shares sum to 100%.

Segment	VaR share	ES share
FLB_short	8.7%	8.8%
FLB_med	4.2%	4.3%
FLB_long	10.0%	10.0%
Corp_L1	14.9%	14.9%
Corp_L2	10.2%	10.3%
Corp_L3	28.3%	28.2%
MBS	23.6%	23.5%

Unconditionally, risk is concentrated in the credit-sensitive segments: third-listing-tier corporate issues (Corp_L3) contribute about 28% of portfolio VaR, and mortgage bonds (MBS) about 24%, even though their portfolio weight is 14% each. The regime-conditional picture is more informative (figure 8): in the quiet regime R1 nearly all of the tail risk (73%) is concentrated in Corp_L3, whereas in the sell-off regime R2 the contribution redistributes toward MBS (37%) and Corp_L1 (23%). This shows that the source of portfolio risk *changes* with the regime — a

property that unconditional measures conceal.

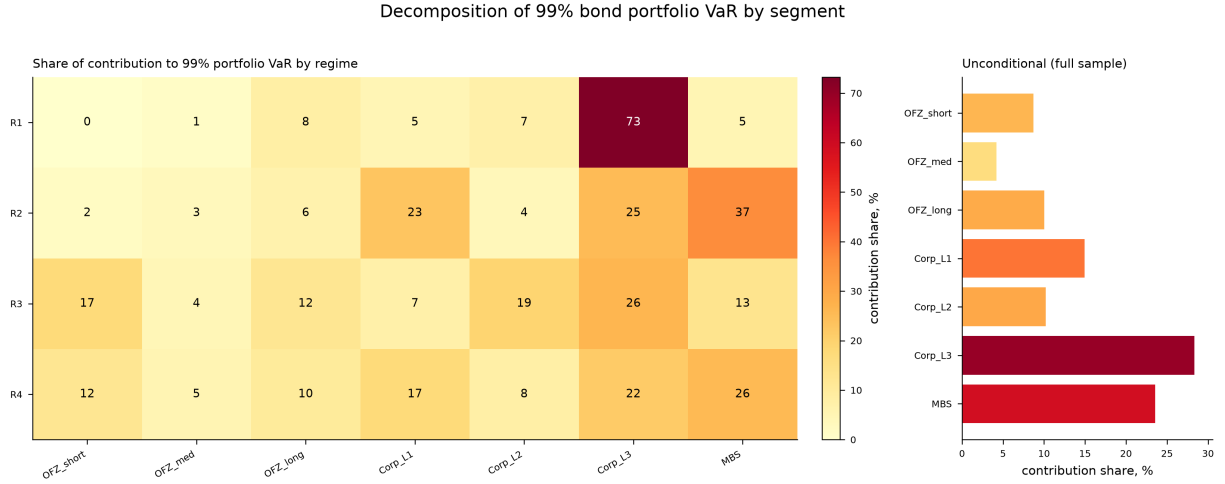


Figure 8: Left: share of each segment in the portfolio's 99% VaR by regime. **Right:** unconditional decomposition over the full sample. Equally-weighted portfolio.

9 Bootstrap: Confidence Intervals

Point estimates do not reflect parameter uncertainty on a sample of 904 days. We apply the stationary block bootstrap (Politis–Romano, expected block length ≈ 20 days, preserving autocorrelation and volatility clustering) and refit the t-HMM on each resample ($B = 163$ successful ergodic fits out of 200), matching regimes by portfolio volatility.

Table 10: Point estimates and 95% percentile bootstrap intervals. Degrees of freedom ν of regimes R3/R4 are weakly identified (the upper bound is capped at 200).

Parameter	Estimate	95% CI
ν R1	2.88	[2.65, 63.18]
ν R2	5.19	[2.50, 45.99]
ν R3	14.18	[3.51, 200.00]
ν R4	7.94	[3.49, 200.00]
Vol. R1, %	2.96	[1.54, 3.84]
Vol. R2, %	3.80	[2.99, 4.65]
Vol. R3, %	4.01	[3.65, 5.73]
Vol. R4, %	6.18	[4.80, 8.56]
Unconditional VaR 95%	-0.41	[-0.44, -0.34]
Unconditional ES 95%	-0.60	[-0.64, -0.50]
Unconditional VaR 99%	-0.70	[-0.76, -0.58]
Unconditional ES 99%	-0.91	[-1.04, -0.75]
Breach frequency 99%, %	0.88	[0.55, 1.99]

Regime volatilities and unconditional VaR/ES are estimated precisely (narrow intervals), whereas the degrees of freedom of the large- ν regimes (R3: $\nu = 14$, CI up to 200; R4: $\nu = 8$, CI up to 200) are weakly identified — the data confidently distinguish "heavy tails" from "nearly Gaussian," but do not pin down a specific large value of ν (Figure 9). In contrast, the heavy tails

of the quiet regime R1 ($\nu = 2.9$, CI [2.6, 63]) and the sell-off regime R2 ($\nu = 5.2$, CI [2.5, 46]) are robust. This is an honest characterization of what the sample can and cannot tell us.

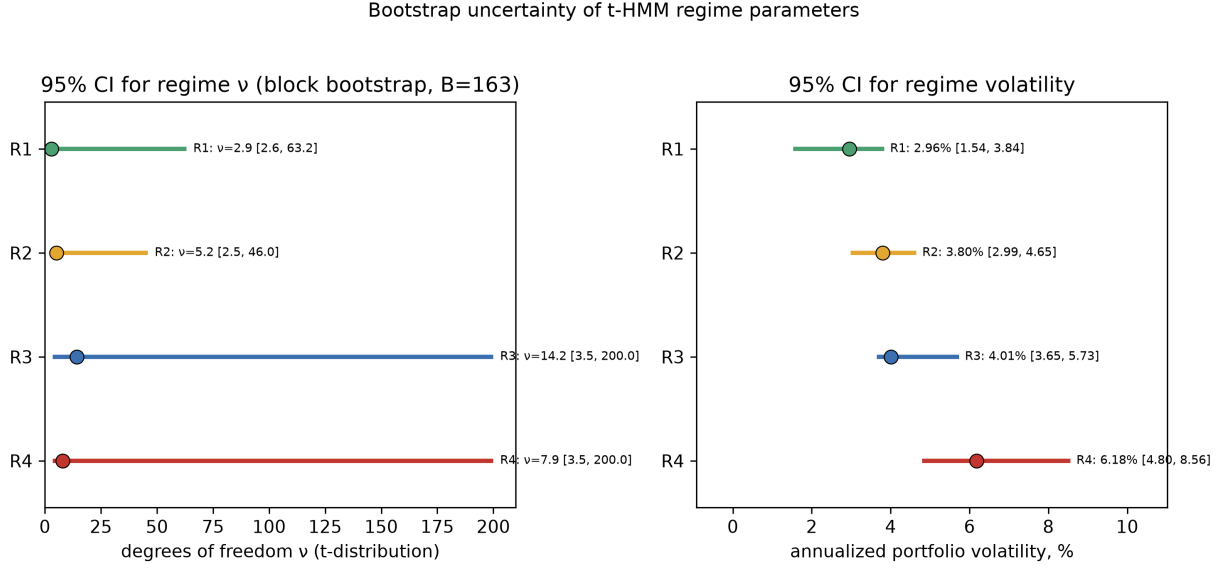


Figure 9: 95% bootstrap intervals. **Left:** degrees of freedom ν by regime. **Right:** annualized volatility by regime.

10 Multivariate benchmark: DCC-GARCH

EVT-GPD and GARCH- t are univariate and operate on the already aggregated portfolio. A fair multivariate competitor to t-HMM is the dynamic conditional correlation model DCC-GARCH (Engle, 2002): for each of the 7 segments a GARCH(1,1) with Student innovations is estimated, and the correlation matrix evolves according to the DCC recursion $\mathbf{Q}_t = (1 - a - b)\bar{\mathbf{Q}} + a\epsilon_{t-1}\epsilon_{t-1}^\top + b\mathbf{Q}_{t-1}$. The portfolio VaR is obtained from the forecast of the full covariance matrix $\mathbf{H}_t$. The model is estimated on the same 403 out-of-sample days.

Table 11: VaR coverage backtest: six models over 403 out-of-sample days (with DCC-GARCH added).

Level	Model	Breaches	Share	Kupiec p	Christoff. CC p	Verdict
95%	t-HMM	20	4.96%	0.973	0.999	—
95%	Gaussian	24	5.96%	0.392	0.652	—
95%	Empirical	26	6.45%	0.200	0.377	—
95%	EVT-GPD	27	6.70%	0.136	0.266	—
95%	GARCH- t	19	4.71%	0.791	0.959	—
95%	DCC-GARCH	12	2.98%	0.045	0.092	rejected
99%	t-HMM	2	0.50%	0.260	0.525	—
99%	Gaussian	10	2.48%	0.012	0.034	rejected
99%	Empirical	5	1.24%	0.640	0.842	—
99%	EVT-GPD	4	0.99%	0.988	0.960	—
99%	GARCH- t	2	0.50%	0.260	0.525	—
99%	DCC-GARCH	2	0.50%	0.260	0.525	—

DCC-GARCH delivers calibrated coverage at the 99% level (0.50%, Kupiec $p = 0.260$) — on par with t-HMM — but at the 95% level it is **rejected** ($p = 0.045$): its VaR there is too

conservative (2.98% breaches versus the 5% nominal). The average pairwise segment correlation estimated by DCC rises from 0.12 to 0.28 in stress windows (figure 10, right) — the model captures the increase in co-movement, but through a single correlation dynamic, without tail switching. The key difference: both t-HMM and DCC are calibrated in the far tail, but DCC does not provide interpretable regimes, regime-switching probabilities, or a regime-conditional risk decomposition — the economically meaningful picture that t-HMM provides.

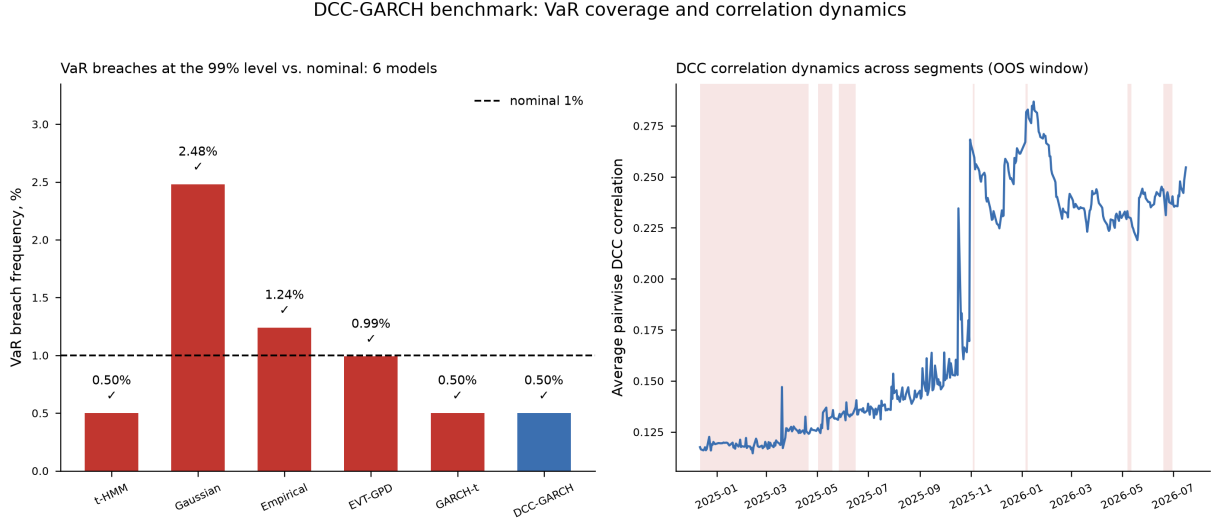


Figure 10: Left: 99% VaR breach frequency across six models. **Right:** dynamics of the average pairwise DCC correlation across segments; shading marks realized stress episodes.

11 Macro-dependent transitions and early warning

Until now, the transition matrix was constant — regimes were described post hoc. To link regime switching to the macro environment, we estimate a reduced-form multinomial logit for the probability of transition as a function of macro covariates (the CBR key rate, inflation, USD/RUB volatility and return, rate change), controlling for the originating regime. *This is a reduced-form analysis on a fixed decoded path, not a full EM re-estimation of a nonstationary HMM.*

Table 12: Logit coefficients for entry into stress regimes (R2UR4), pooled with fixed effects for the originating regime. Standardized covariates; * denotes $p < 0.05$.

Factor	Coeff.	Odds ratio	p
CBR key rate	+0.794	2.21*	0.005
Inflation	+0.001	1.00	0.997
USD/RUB vol.	-0.258	0.77	0.274
USD/RUB, 20-day return	-0.019	0.98	0.901
Rate change over 60 days	+0.042	1.04	0.809

The main and only significant driver of entry into stress is **the level of the CBR key rate** (odds ratio 2.21 per standard deviation, $p = 0.005$): at a high rate, the probability of transitioning into a sell-off or high-volatility regime doubles. Inflation and exchange-rate variables are insignificant after controlling for the rate (figure 11, bottom). The daily signal constructed from the model — the forecast probability of being in a stress regime tomorrow — separates realized stress episodes with an out-of-sample AUC = 0.88, i.e., it works as a practical early-warning

indicator (figure 11, top).

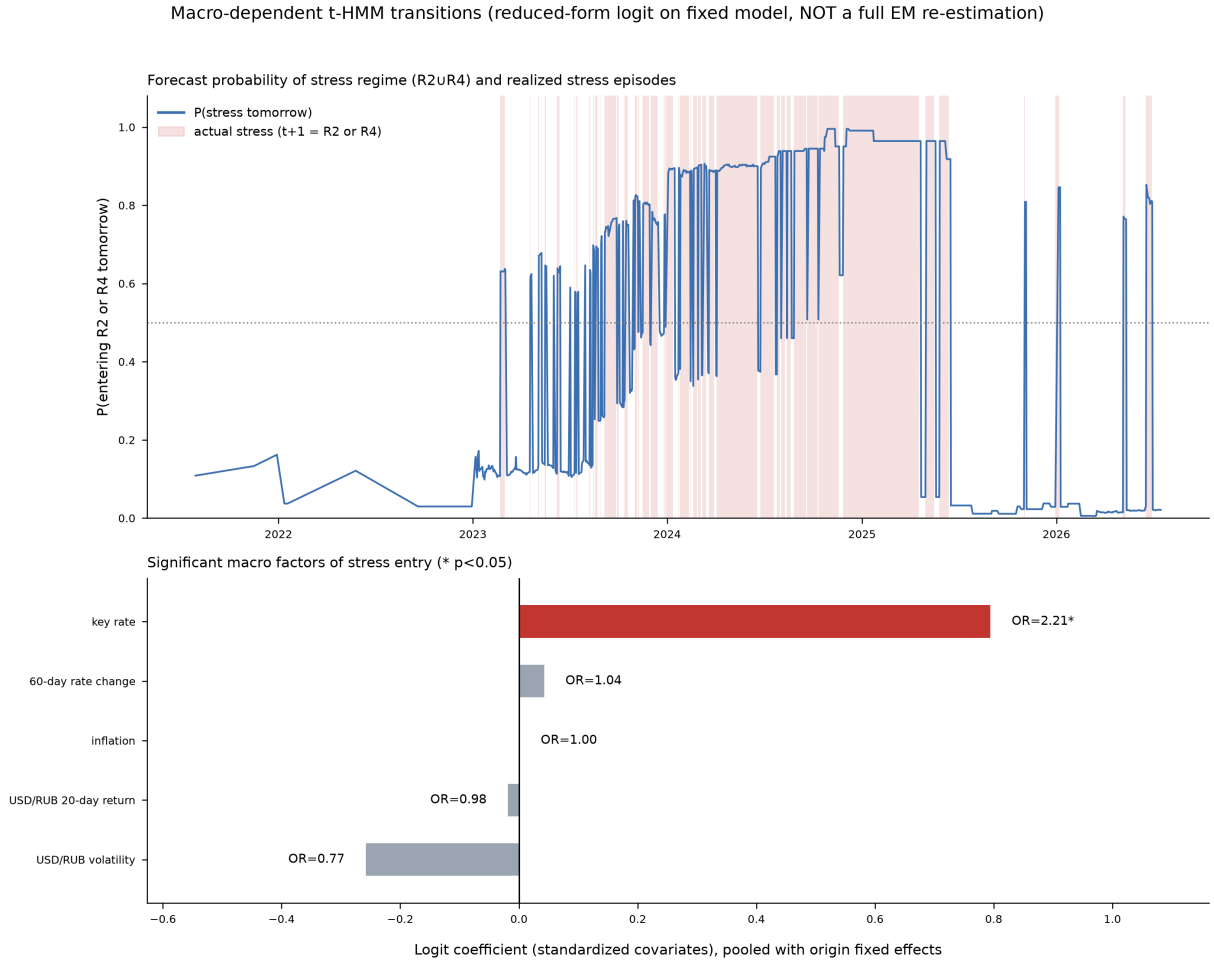


Figure 11: Top: forecast probability of the stress regime (R2UR4) for the next day; shading — actual stress episodes. **Bottom:** logit coefficients of macro factors for entry into stress.

12 Limitations

- Risk measures and the Euler decomposition are given for an equally-weighted portfolio of segments; the bank's actual weights change the absolute values and segment contributions (not the regime structure) — the decomposition formula accommodates an arbitrary weight vector.
- Coupon income is accounted for as smoothed carry at the annual rate; precise accounting for coupon payment dates and reinvestment would yield a finer intra-month structure, but has little effect on daily volatility and regimes.
- Eurobonds in the portfolio (16 issues) are excluded due to their absence on MOEX; full coverage would require an external source (e.g., Cbonds).
- The comparison includes univariate (EVT-GPD, GARCH- t) and multivariate (DCC-GARCH) benchmarks; a full Markov-switching GARCH (MS-GARCH), combining GARCH dynamics with regime switching, is a natural further extension.
- The analysis of macro-dependent transitions is carried out in reduced form on a fixed decoded regime path; a full EM estimation of a nonstationary HMM with a macro-driven transition

matrix is a more rigorous but substantially more costly alternative.

- Estimates of large ν (regimes R3, R4) are weakly identified (wide bootstrap intervals); the conclusion «tails are close to normal» is reliable, but the specific value of ν is not.

13 Conclusion

The multivariate Student HMM identifies four economically interpretable regimes of bond portfolio returns and statistically dominates the Gaussian analogue. Tail heaviness, measured by the degrees of freedom ν , varies substantially across regimes. A rigorous rolling out-of-sample back-test confirms that accounting for heavy tails is essential for correct risk estimation at the 99% level: the normal approximation is rejected, whereas the t-HMM is calibrated and performs no worse than dedicated tail models, while additionally providing a regime picture unavailable to univariate alternatives. Extended analysis confirms and deepens this conclusion: the Expected Shortfall of the normal approximation also statistically underestimates risk, whereas the t-HMM is calibrated; the multivariate DCC-GARCH achieves correct coverage only in the far tail, without providing interpretable regimes; bootstrap analysis shows that the heavy tails of the calm and sell-off regimes are estimated robustly, while high ν values are identified only as “close to normal”; Euler risk decomposition attributes the bulk of tail risk to the credit segments (Corp_L3, MBS) and shows its redistribution across regimes; finally, the macro-dependent transition analysis identifies the level of the Central Bank policy rate as the main driver of entry into stress and provides a working early-warning signal. The approach transfers directly to the bank’s actual portfolio weights (via substitution of the weight vector into the risk decomposition) and to Cbonds data under the same methodology.

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